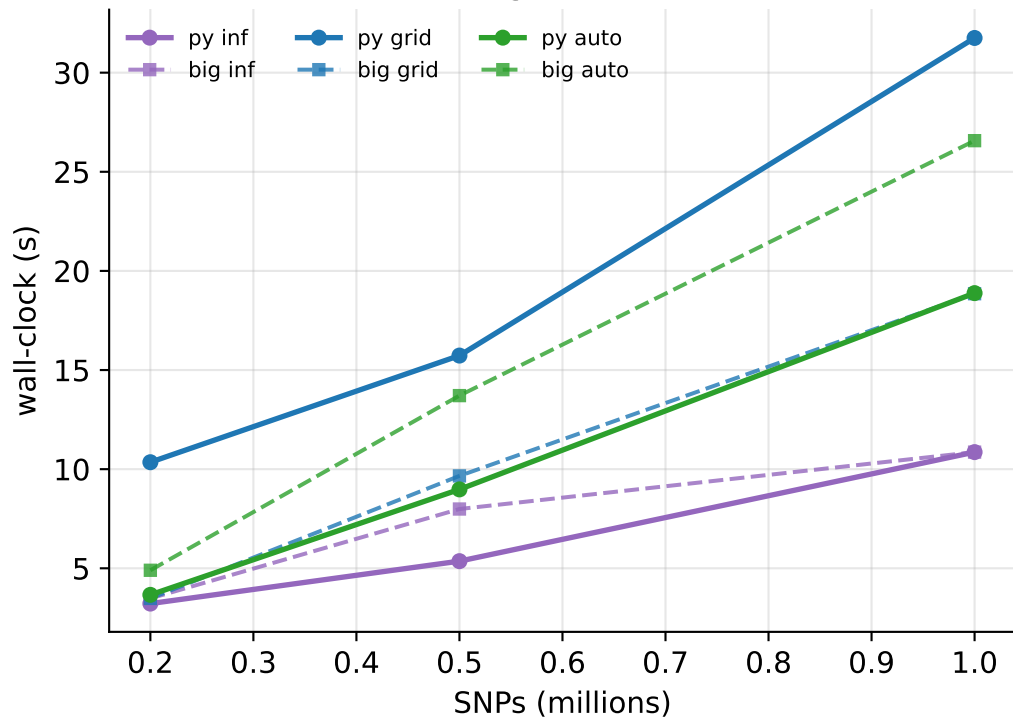
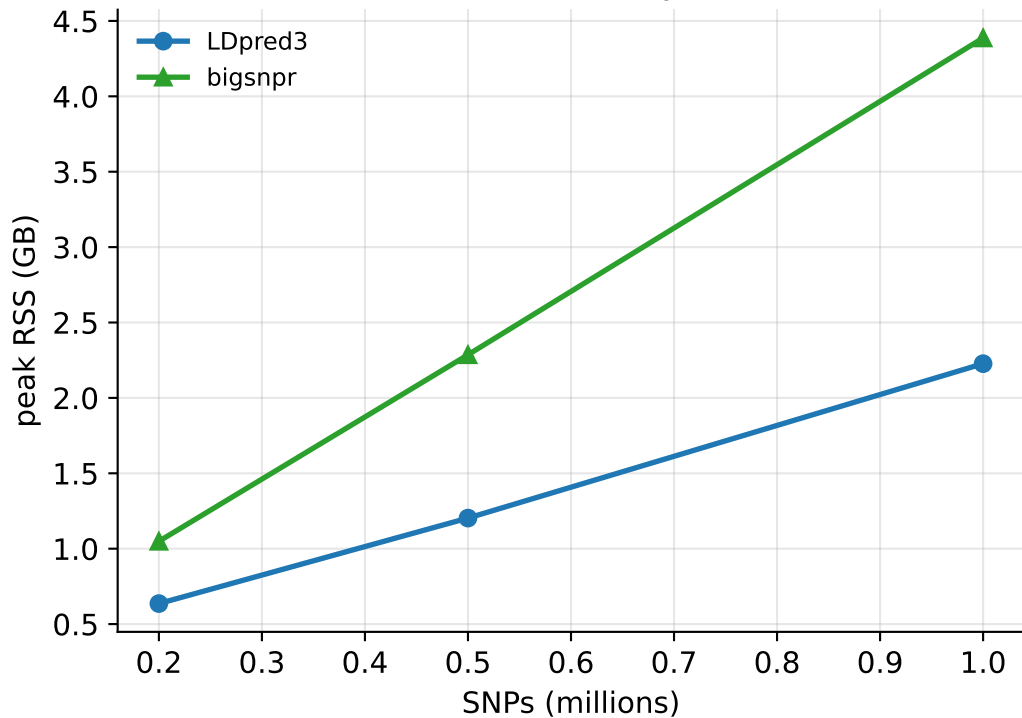


LDpred3 vs bigsnpr — realistic LD (coalescent), N=50k, $h^2=0.5$ (auto: oracle init, both tools)

Running time, 1 core

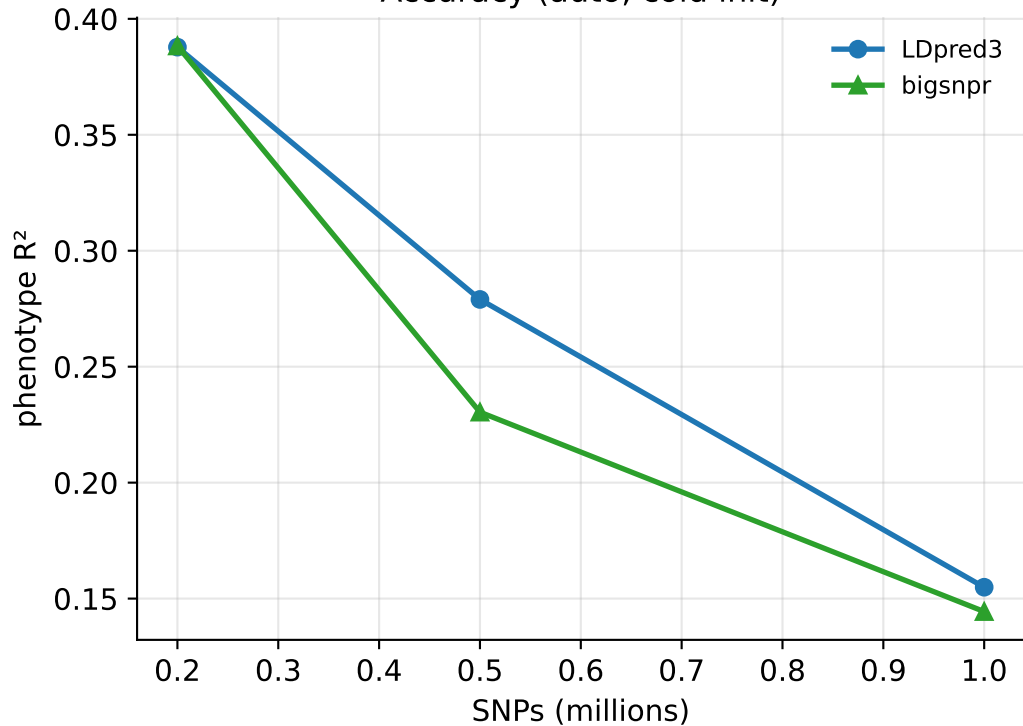


Peak memory

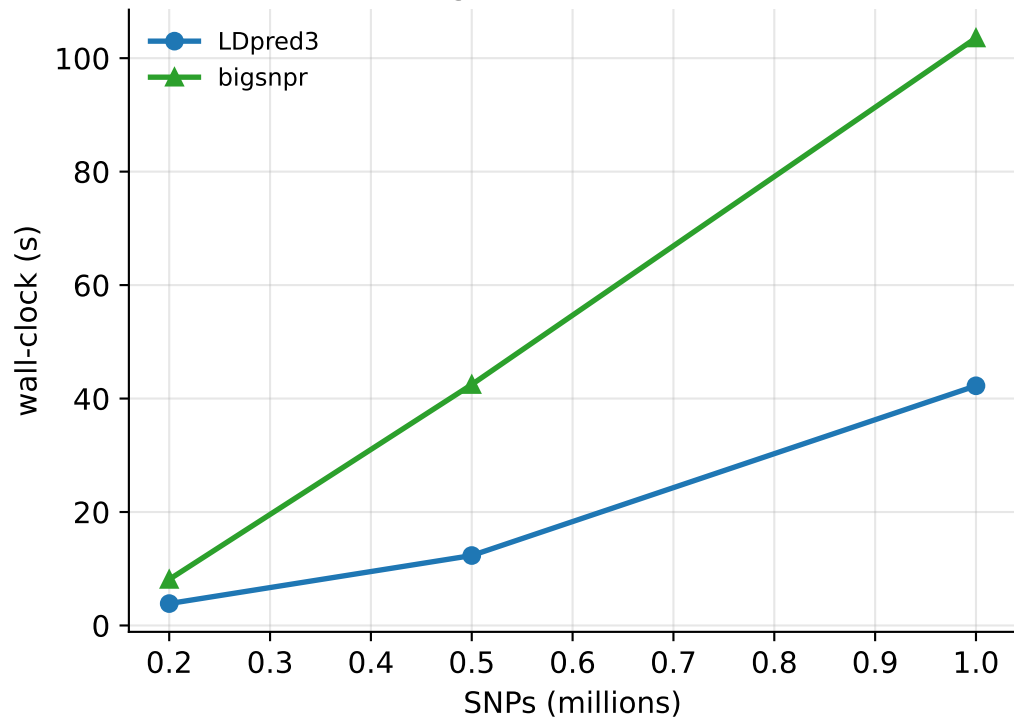


auto from a cold start ($h^2_{\text{init}}=0.1$, $p_{\text{init}}=0.1$, both tools, single chain)

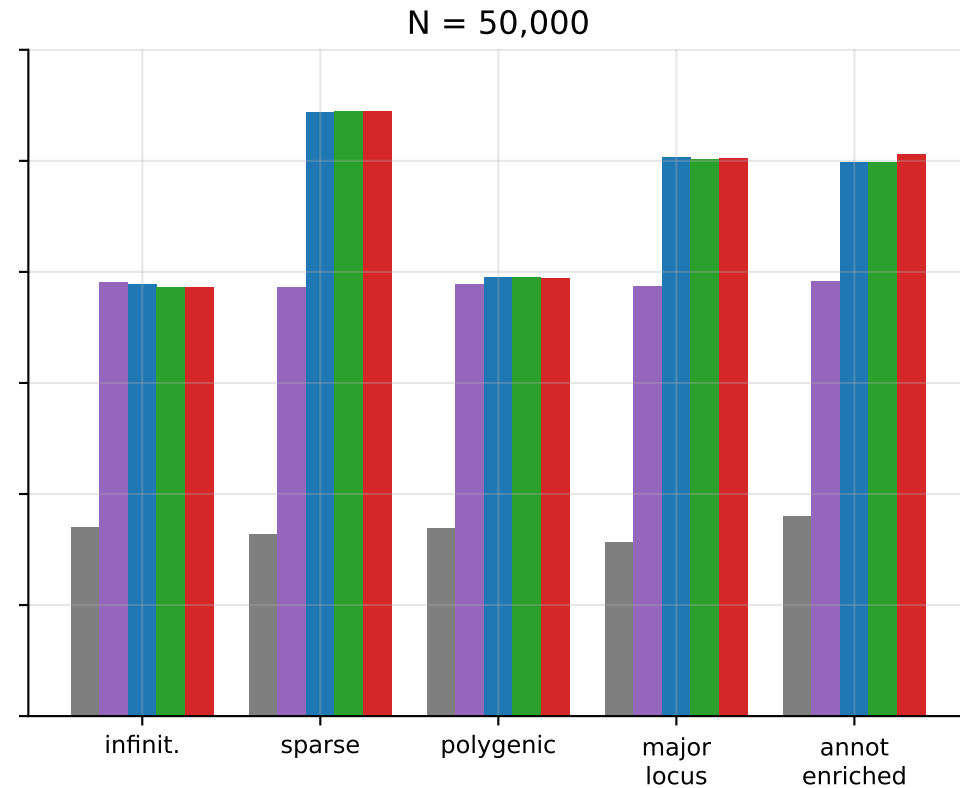
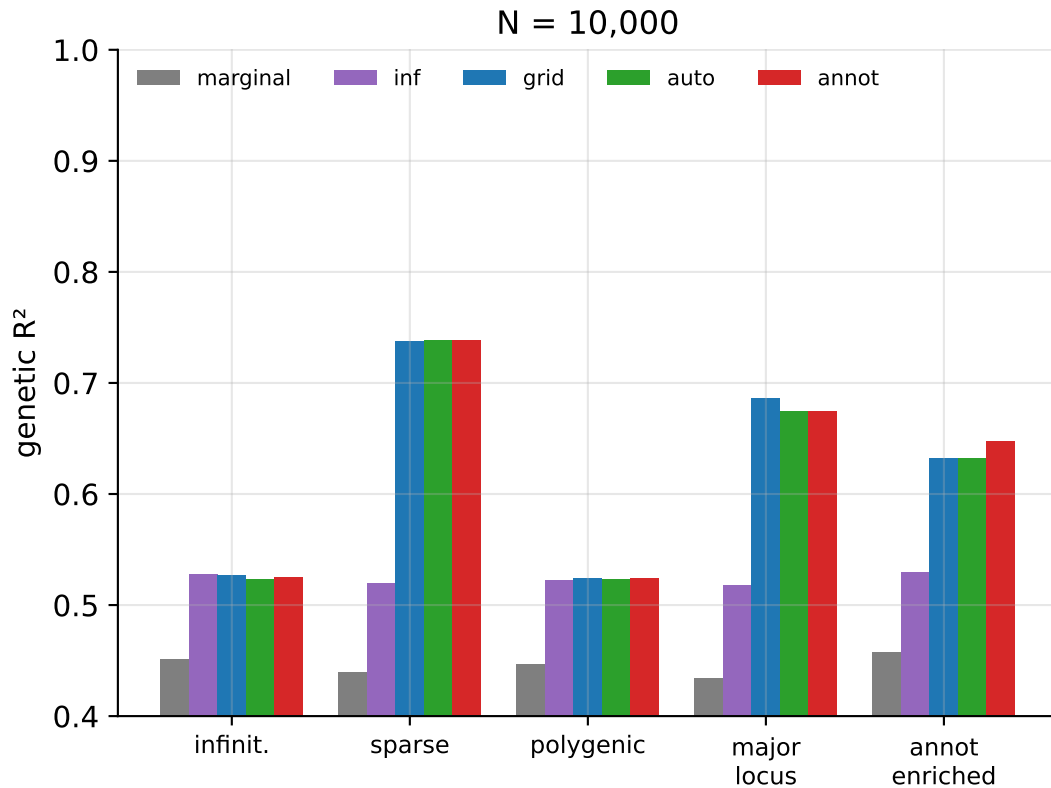
Accuracy (auto, cold init)



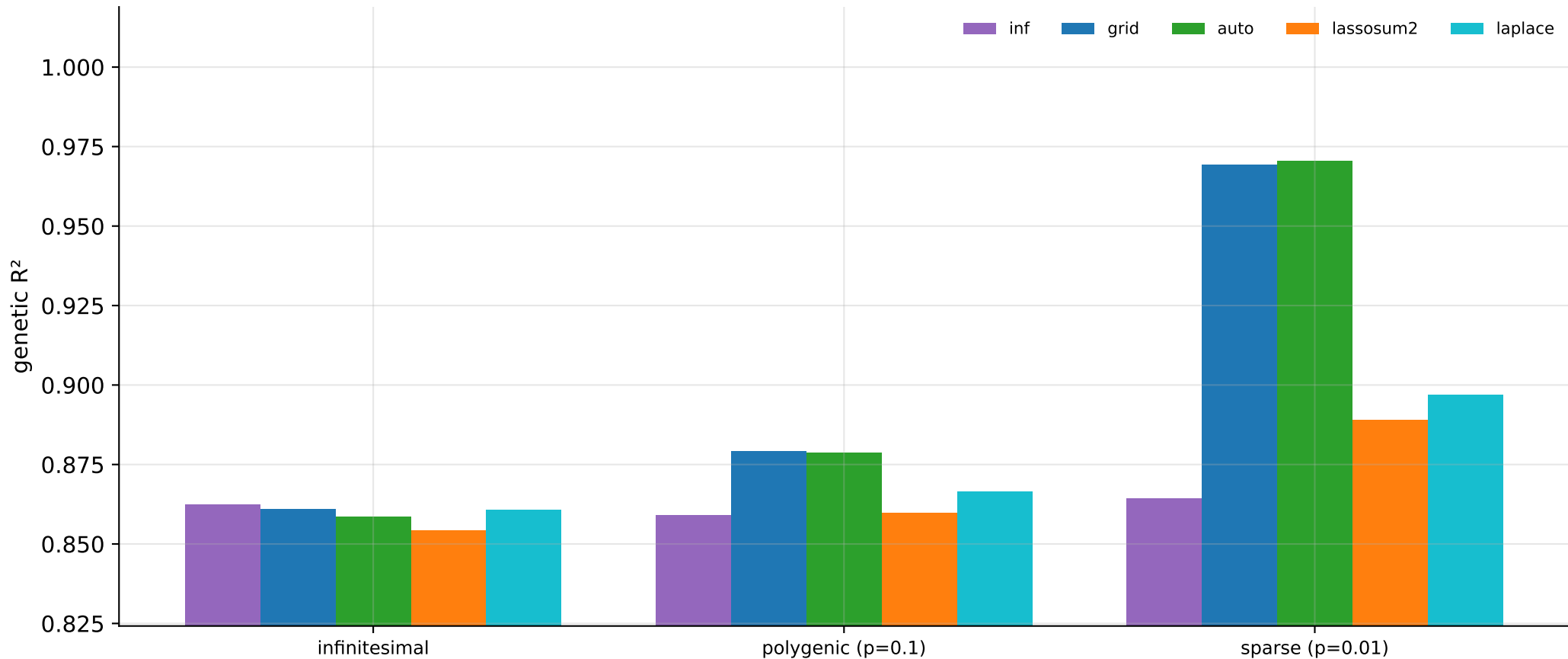
Running time (auto, cold init)



Accuracy by genetic architecture (realistic LD, $h^2=0.5$)

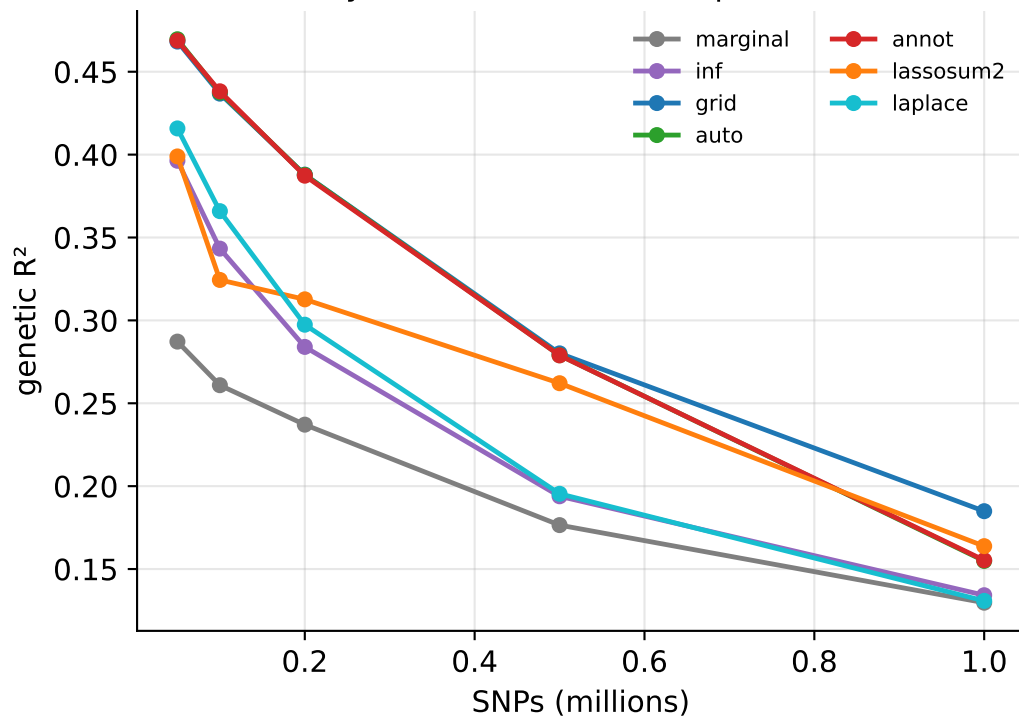


Sparse-prior methods by architecture — Bayesian lasso (laplace) vs the lasso (lassosum2)

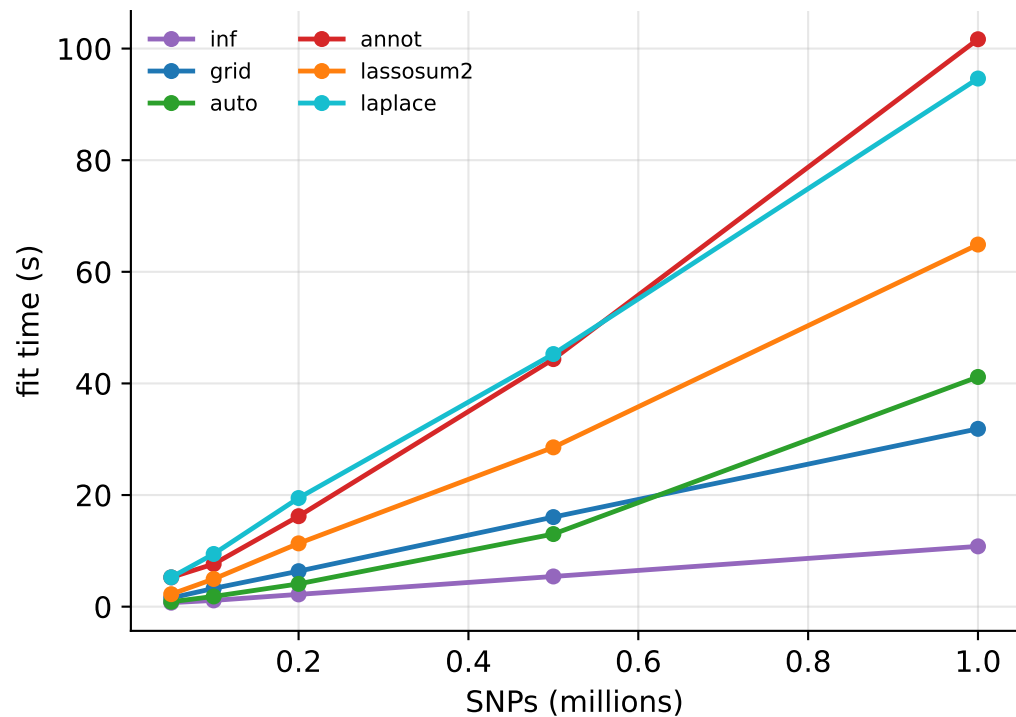


Methods: accuracy & running time at scale (realistic LD)

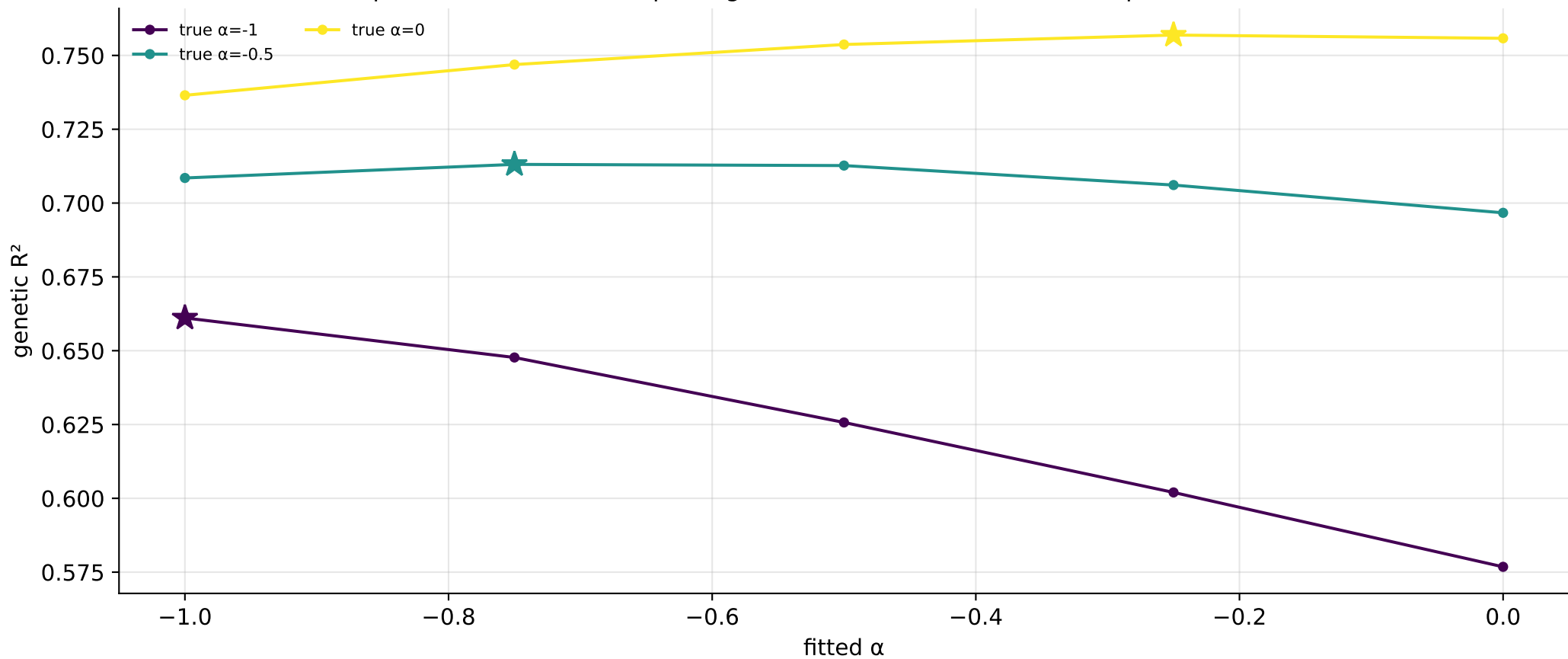
Accuracy vs #SNPs (fixed N — power dilutes)



Fit time vs #SNPs (1 core)

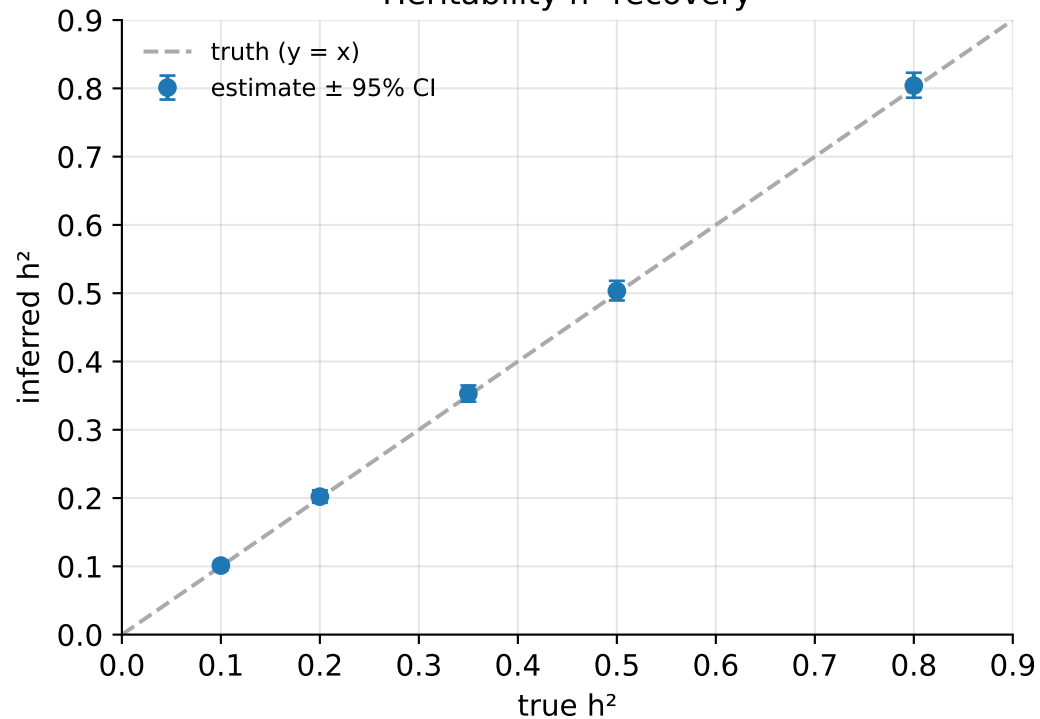


MAF-dependent slab-variance prior: genetic R^2 vs fitted α (★ = best; peaks track the true α)

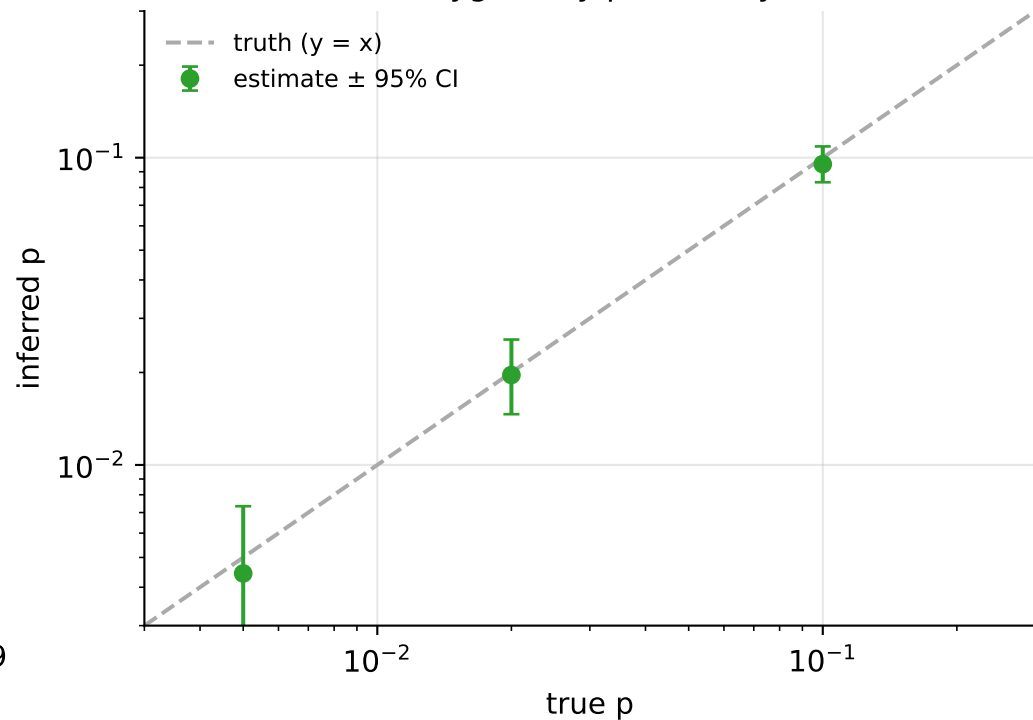


Inference evaluation (LDpred3-auto-infer, no validation cohort)

Heritability h^2 recovery

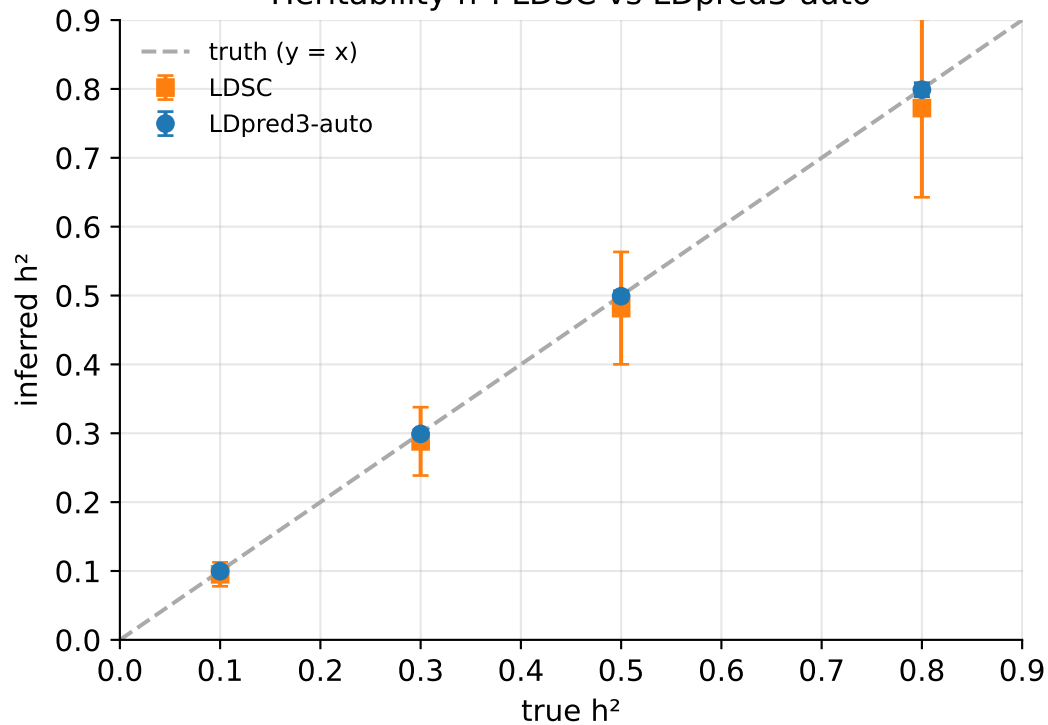


Polygenicity p recovery

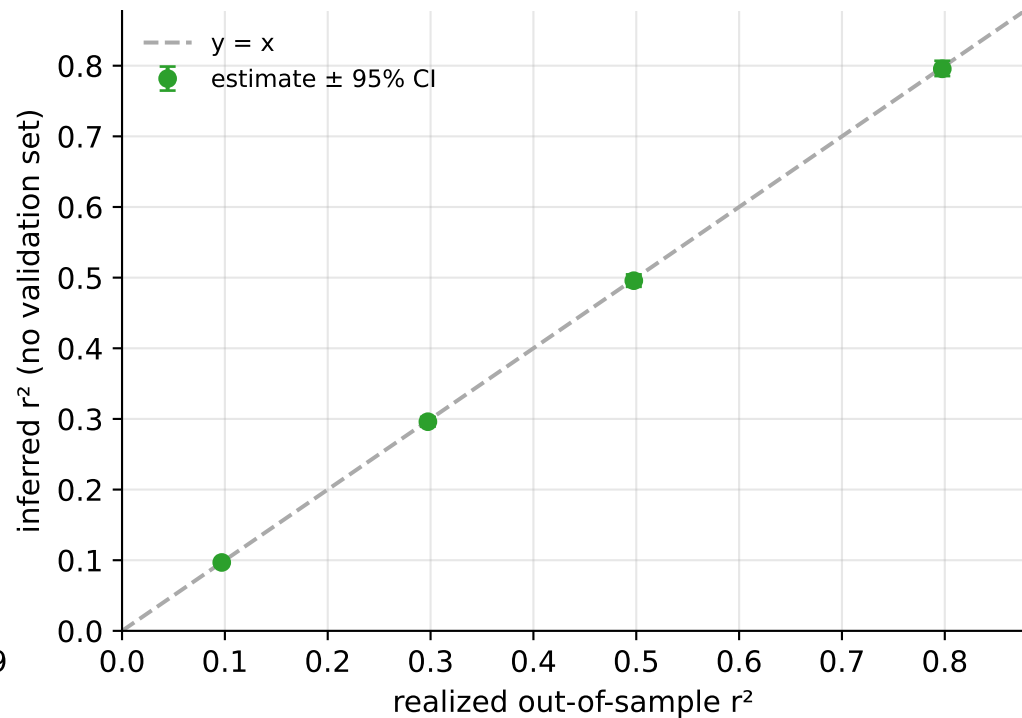


Inference cross-checks (realistic coalescent LD)

Heritability h^2 : LDSC vs LDpred3-auto

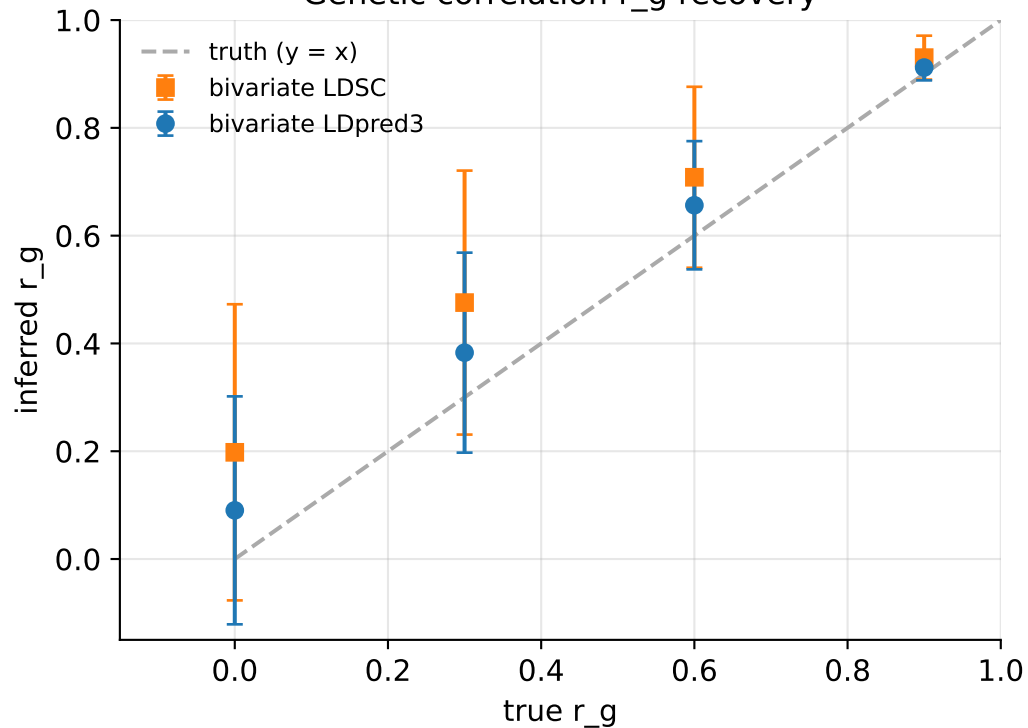


Predictive r^2 : estimated vs realized

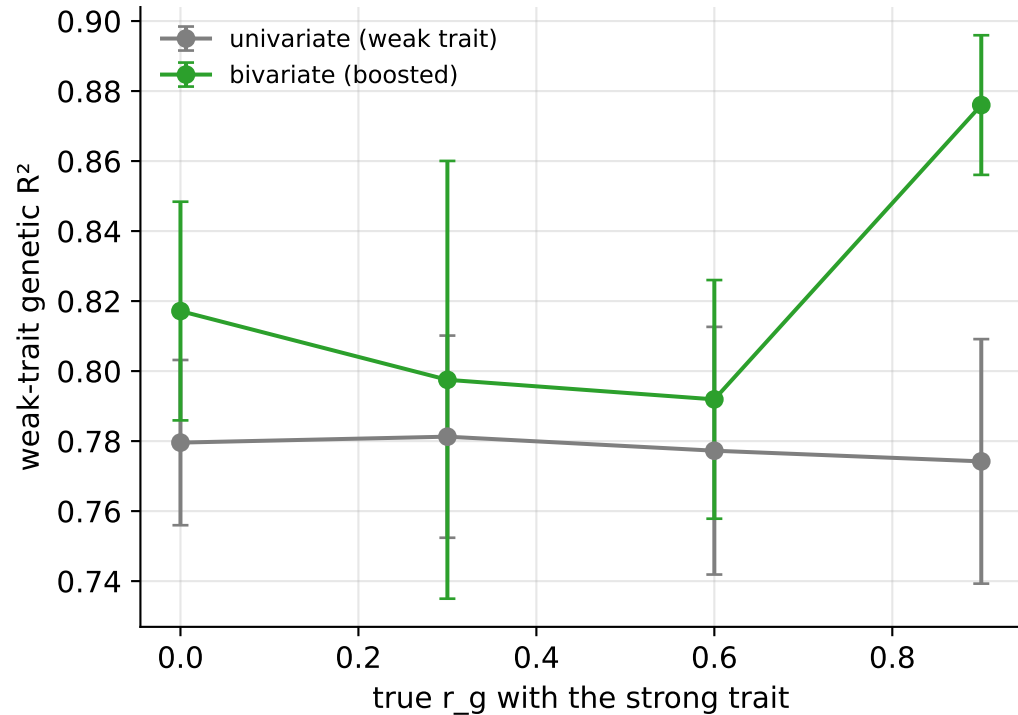


Bivariate analysis (realistic coalescent LD)

Genetic correlation r_g recovery

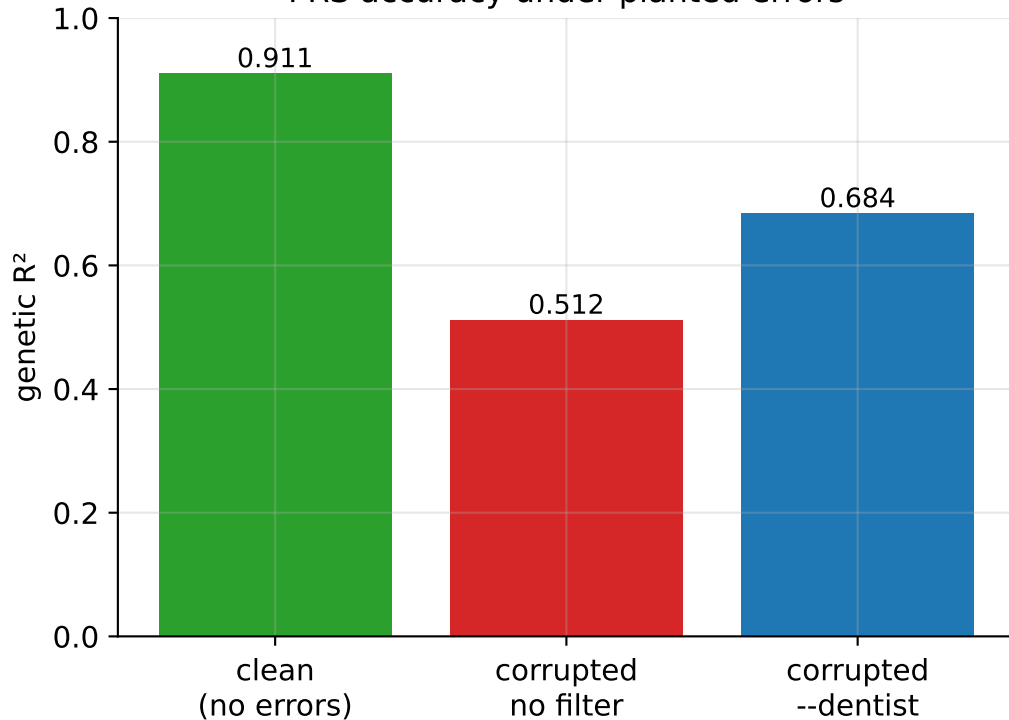


Weak-trait prediction gain ($N_2 \ll N_1$)

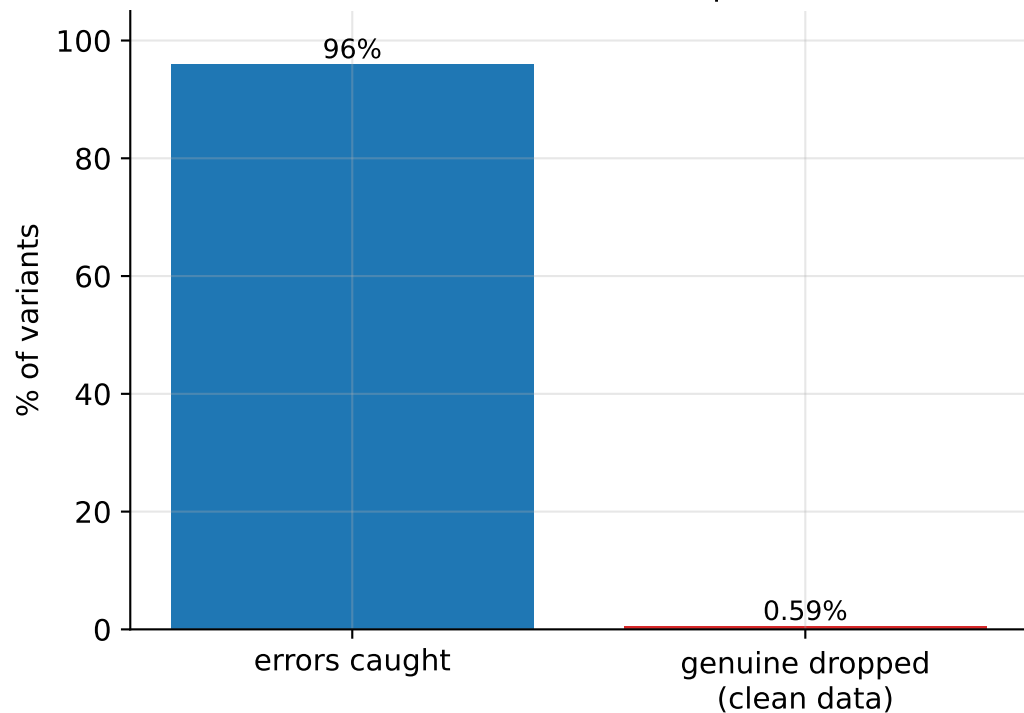


DENTIST LD-consistency filter (--dentist)

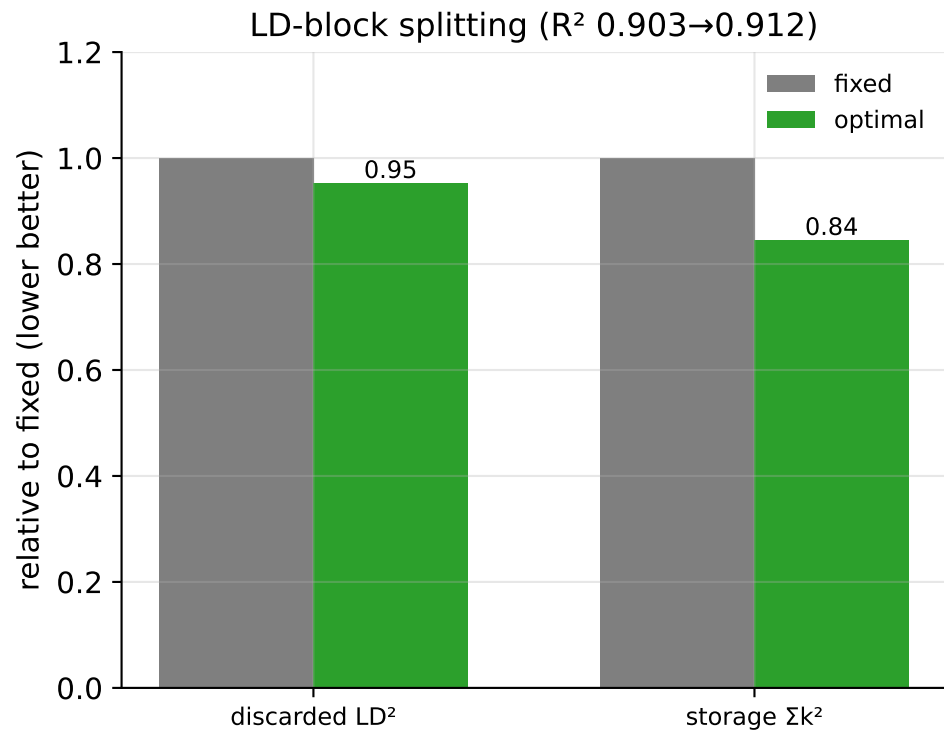
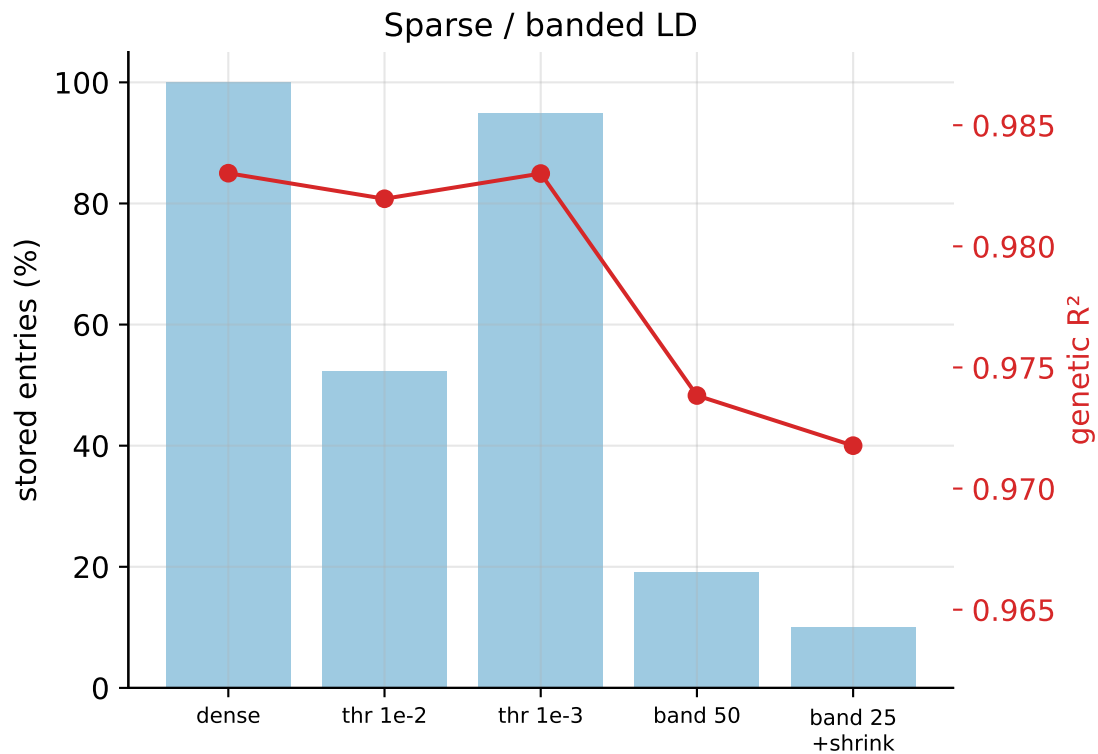
PRS accuracy under planted errors



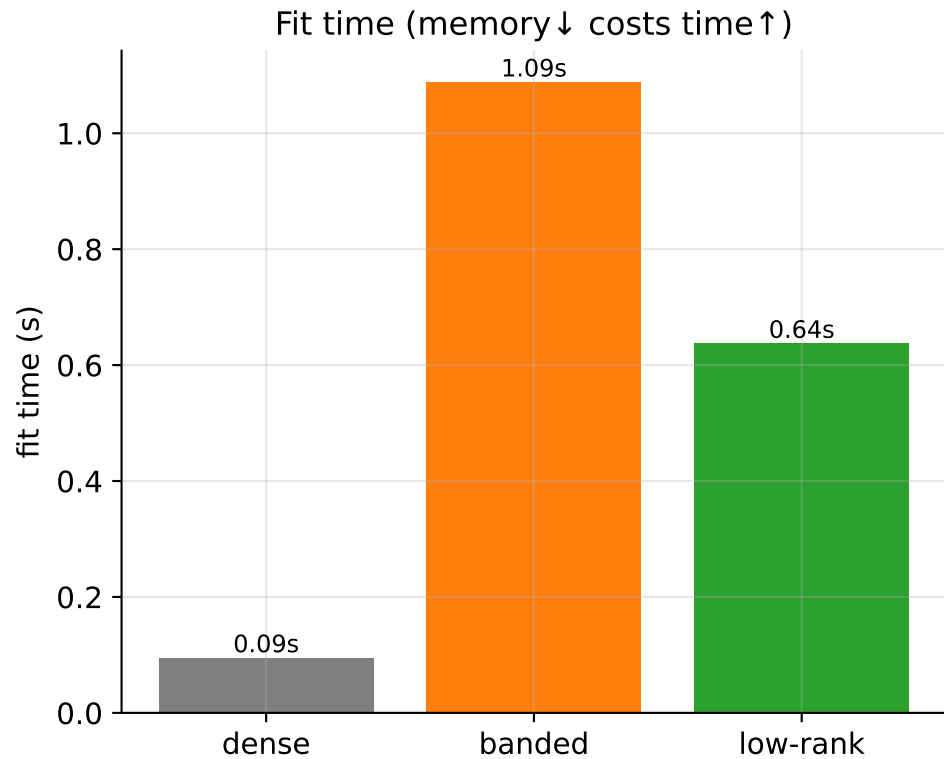
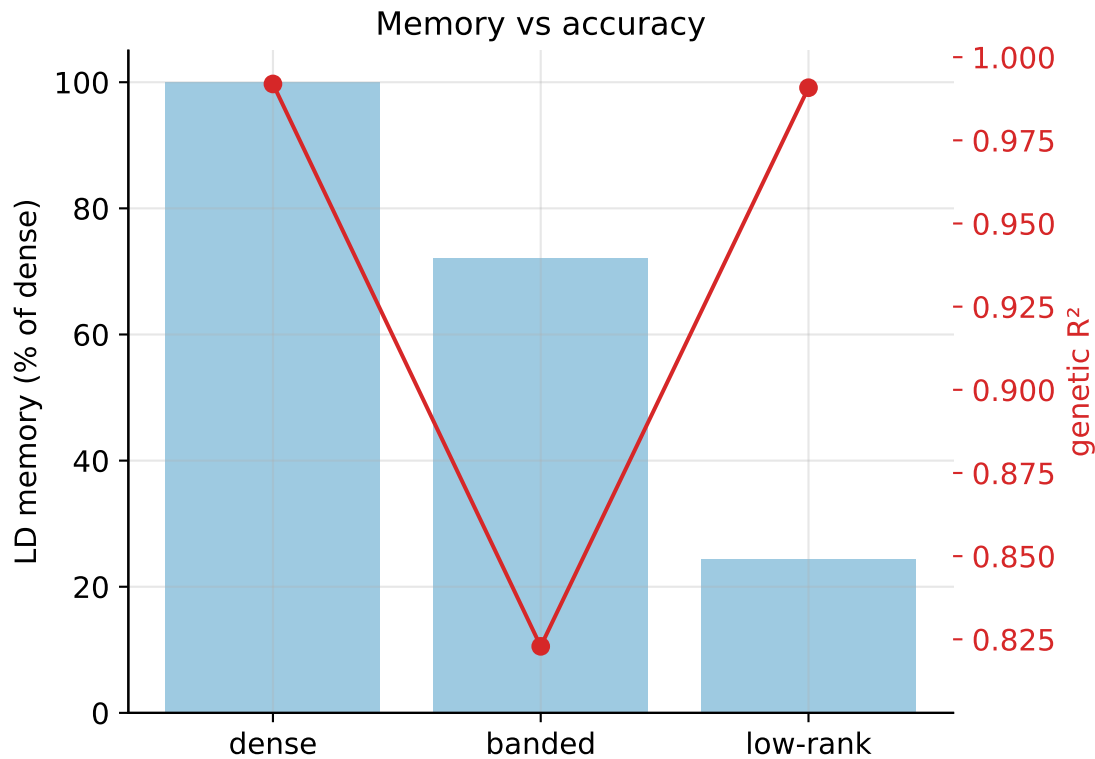
Catch rate vs false-drop cost



LD representation: storage vs accuracy



LD representations at scale — realistic coalescent LD (low-rank matches dense at ~1/4 memory; banding loses accuracy)



Performance

